

Are Big Cities Important for Economic Growth?*

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ABSTRACT: Cities are often described as engines of economic growth. We assess this statement quantitatively. We focus on two mechanisms: a static agglomeration effect that makes production in bigger cities more efficient, and a dynamic effect whereby urban scale impacts the productivity of invention, which in turn determines the speed of technological progress for the country as a whole. Using estimates of these effects from the literature and MSA-level patent and population data since 1900, we ask how much lower US output would be in 2010 if city size had been limited to one million or one hundred thousand starting in 1900. These effects are small. If city sizes had been limited to one million people since 1900, output in 2010 would have been only 8% lower than its observed value.

JEL: O40, R10

Keywords: Agglomeration economies, Economic growth

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1. Introduction

Cities are economic dynamos. They are hubs of innovation and breeding grounds for new industries. Workers, entrepreneurs, and scientists congregate in cities to take advantage of thick markets and information spillovers. In the US in 2010, the 27 cities that constituted the top decile of the MSA size distribution accounted for 47% of US population, 54% of output, and 65% of patents (the data are discussed below). Cities are often referred to as “engines of economic growth,” and thinkers including Marshall, Jacobs, Lucas, and Romer argue for their importance in the process of economic growth.

In this paper we evaluate this idea quantitatively. Specifically, we assess the contribution of big cities to growth in the United States by comparing observed US income growth between 1900 and 2010 to counterfactual paths of growth in scenarios where city sizes are limited to either one million or 100,000 people.

At the core of our exercise is an attempt to quantify the channels by which city size affects both static productivity and productivity growth over time. The latter, we stress, is a result of technological progress that diffuses from the city in which new technologies are created to the country as a whole. Much of the existing literature on urban structure and growth restricts attention to static effects; long-run productivity growth is either taken as exogenous or related via complex and somewhat speculative channels to higher urban productivity causing higher investment in human capital. While there exists a literature on the effect of urban structure on R&D, the full linkage from urban structure to research to long run TFP growth has not previously been part of the analysis of how cities affect growth.

We adopt the growth accounting literature’s convention of apportioning growth into shares due to increased total factor productivity (TFP) and accumulation of factors of production. The overwhelming conclusion of this literature is that measured TFP growth has been the most important contributor to growth of output per worker in the United States. For example, using data from Gordon (2016a), the average annual growth rate of output per hour in the United States over the period 1890-2014 was 2.12%. Of that 1.57% per year, or about 76% of the total, was attributable to TFP growth, with increased observable human capital per worker explaining the bulk of the remainder.¹

This suggests that, if limiting city sizes would have caused a big decrease in the growth rate, this would be because limiting city sizes decreased growth of TFP. We consider four mechanisms through which this might have occurred,

¹Specifically, measured TFP grew at a rate of 1.10% over this period (Gordon (2016b)). We follow Jones (2002) in attributing growth in physical capital that does not change the capital/output ratio to TFP rather than capital deepening. Growth due to TFP is thus $1/(1 - \alpha)$ times as large as TFP growth, where α is capital’s share of income, for which we use a value of 0.3.

- *Urban scale effects on output.* A large literature emphasizes productivity gains that result from larger city sizes. Mechanisms include thicker markets, knowledge spillovers, increased specialization, and increased competition. Given the observed increase in the fraction of the population living in large cities as well as the increasing size of these cities, urban scale effects would have produced a rise in TFP during our study period.
- *Urban scale effects on the growth of unobserved human capital.* Holding observed measures of education constant, wages rise more quickly for people who start their careers in bigger cities, and on average, their higher wages follow them if they later migrate to smaller cities. These facts suggest that working in a large city increases individual human capital. However, conventional productivity measures usually ignore this effect, so that the extra human capital growth that is due to cities ends up as part of measured TFP growth. In the interests of brevity, we often abuse language and refer to this mechanism as “urban scale effects on human capital.”
- *Urban scale effects on innovation.* Inventive activity is concentrated in big cities, and inventors are more productive in these. A version of the US that had smaller cities would have *ceteris paribus* had less technological progress and slower growth of TFP.
- *Improved Sorting.* If cities differ in their levels of productivity for intrinsic reasons unrelated to their size, then the aggregate level of TFP in the country will be a function of how efficiently people are allocated to high-productivity cities. If the efficiency of this allocation improves over time, it will cause aggregate TFP to increase. The existence of large cities may allow this sorting to be more efficient by allowing more people to live in any given high-productivity location.

We discuss each mechanism at length in section 3.

In the body of the paper we use a simple growth model featuring urban scale effects on output, the growth of unobserved human capital, and innovation. The main equations of the model are variants of well studied regression equations relating the levels of output, unobserved human capital accumulation, and research production (all considered as dependent variables) to city size, along with one equation relating TFP growth to national R&D.

The fact that the main model equations correspond to well studied regressions means that calibrating the model is simple. We rely on parameter estimates taken from well known estimates of our model’s main equations, and then apply the model to data describing city level output, patenting, and population for the US from 1900 to 2010. By comparing aggregate growth in the observed baseline to counterfactuals with exogenous restrictions on city sizes, we assess the importance of large cities as engines of growth.

The structure of our model, along with the particular structure of the counterfactuals that we construct, allows our analysis to proceed on the basis of remarkably weak assumptions and produces a transparent mapping from data and parameter estimates to results.

Our main finding is that the existence and growth of large cities accounts for only a small fraction of aggregate income growth. We obtain this result for two main reasons. The first is that estimates of urban scale effects on output and human capital accumulation are simply too small for the growth of city populations to have contributed much to TFP growth over the period we study. For urban scale effects on output or human capital to be quantitatively important, scale effects need to be larger or cities need to grow much faster than what we observe.

The second reason is more subtle. Estimates of urban scale effects on innovation are larger than for output or human capital; large cities allow more innovation from a given R&D labor force. Thus, a version of the US that had smaller cities would *ceteris paribus* have experienced slower TFP growth at any point in time. However, one factor that would not be constant along such a counterfactual path is the *level* of technology. Recent research (e.g. Bloom et al. (2020)) argues for strong negative effect of the level of TFP on the efficacy of R&D in producing technological progress. Because of this negative feedback, we show, a version of the US that had smaller cities over the 20th century would have had only a slightly lower level of TFP as of 2010.

Our model is not suitable for evaluating the fourth channel listed above, the improved sorting of people into high-productivity locations. The literature review presents our reasons for believing that this channel was not a major contributor to observed income growth over the period that we study. The literature review also discusses other mechanisms by which the distribution of city sizes might affect the growth rate of output, along with our rationale for excluding them from the analysis.

The rest of this paper is organized as follows. In Section 2 we examine the relationship between city size, output, and patent production in contemporary and historical cross sections. Section 3 summarizes the parts of the literature on cities and economic growth that are most relevant for the exercise that we conduct in this paper. Our model of urban scale economies in the production of output and human capital is specified, parameterized, and applied to cross-sectional data for 2010 in Section 4. Section 5 does a similar analysis of urban scale effects on R&D, as measured by patents, while Section 6 explores the mapping from R&D to the speed of technological progress. Section 7 combines the static and dynamic effects just described. Section 8 concludes.

2. A First Look at the Data

To set the stage for our investigation, we present data describing the cross sectional relationship between city population, city output, and city R&D as measured by patents. For our cities, we consider a set of 275 constant boundary MSAs in the continental US, defined to the same boundaries as Duranton and Puga (2023), along with a single rural area that aggregates all non-metropolitan counties.² We construct decadal population data by combining population data in replication files from Duranton and Puga (2023) with 1900-1990 county population data from Forstall and NBER (1995). This results in an MSA by decade panel of population stretching from 1900 to 2010.³

We measure output using the county level output data from the BEA (USDOC/BEA/RD, 2023), and aggregate counties to MSAs. These data are available from 2001. We use the 2010 BEA data to describe GDP in the terminal decade of our analysis.⁴ We rely on the CUSP data (Berkes, 2018), to measure patents. These data report on all patents issued by the US Patent office from 1836 to 2015, along with the year of issue and county of residence for all listed inventors. Using these data, and pro-rating patents with multiple inventors, we construct county-by-year counts of patents. Because MSAs are defined as collections of counties, we can easily aggregate to the MSA level. In the data presented below, “patents in 1900” refers to the average number of patents per year over the period 1900-1909, and similarly for other decades. The 2010 decade of CUSP data reflects only six years of data rather than ten.

Figure 1(a) is a histogram of population, output, and patents across cities for the year 2010. Cities are grouped in deciles by population, and we include an eleventh non-MSA category. The figure shows the importance of large cities. In total, the top decile of cities accounted for 47% of population, 54% of output, and 65% of patents.⁵ Large cities have higher per-capita output and patent production than small cities, on average. Non-metropolitan counties accounted for 19% of population, 14% of output and 6% of patents.

Figures 1(b) and 1(c) repeat the analysis of Figure 1(a) for the years 1900 and 1950, although we have no MSA-level output data for these years. The concentration of patenting in the largest decile of cities is less pronounced in 1900 and 1950 than in 2010. In 1900 there is also a significant over-representation of patents in the second

²Strictly, our MSAs are the set of all CMSAs, MSAs and NECMAs drawn to 2000 boundaries.

³Our sample of MSAs decreases slightly in the early part of our sample. This largely reflects the fact that some states had not yet joined the union and so census data and county boundaries do not exist. For example, Arizona, New Mexico and Oklahoma all joined the US after 1900.

⁴The BEA productivity data does not report for the two counties that make up the Danville, VA MSA, and so the BEA data describes 274 MSAs instead of 275.

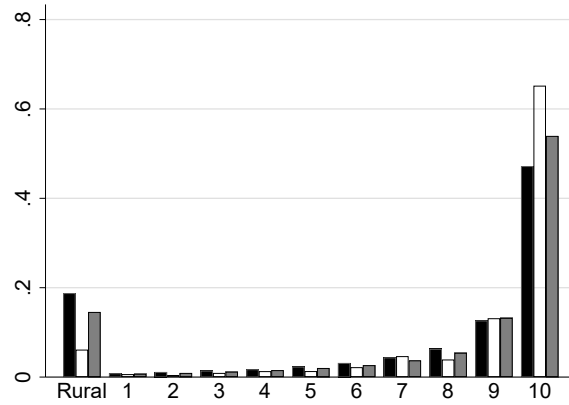
⁵San Antonio, with a population of 1.99 million, is the smallest MSA in the top decile.

largest decile of cities. The under representation of non-MSA areas in patenting is more pronounced in the earlier years.

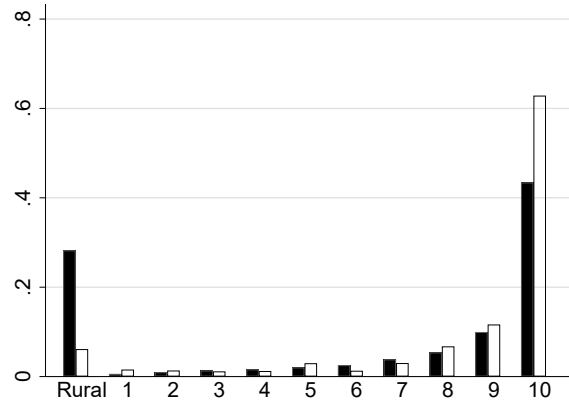
Figure 2 describes correlations in our data in 2010. Panel (a) describes the relationship between the log of MSA output and log population. The tight linear relationship of the logs implies an elasticity of output per capita with respect to population of 8%. This is slightly smaller than the 13% elasticity reported by Glaeser and Gottlieb (2009) for the same regression using data for 2000 and slightly different MSA definitions. Panel (b) describes the relationship between the log of MSA patents and the log of population. The relationship between patents and population is noticeably noisier than is the relationship between output and population, and also considerably steeper. The elasticity of patents per capita with respect to city population in the raw data is 41% in 2010. Panel (c) describes the relationship between patents and output. Unsurprisingly, this relationship is also positive. More interesting is the fact that many smaller cities appear to be relatively specialized in innovation (e.g., Corvallis, Boise, or Rochester, MN) or output, while larger cities almost always do both. San Francisco is an obvious exception. It is the only large MSA that is relatively specialized in innovation.

Figure 3 describes the relationship between patents and city size over time. Panel (a) repeats panel (b) of figure 2. The relationship between patenting and city size is always positive, but flatter as we go back in time. The slope of the regression line is 1.41 in 2010, 1.35 in 1950, and 1.11 in 1900. Looking at the names of cities that are particularly successful at innovation shows the westward march of the center of innovations, from New England in the 19th century, to the Midwest in the 20th century, to the West in the 21st century.

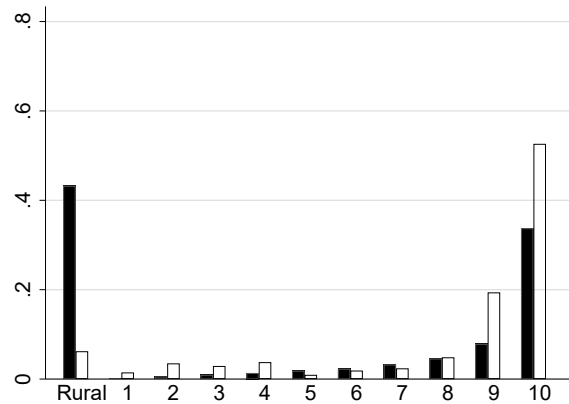
Figure 1: Distribution of population, output and patents by city size in 1900, 1950 and 2010



(a) 2010



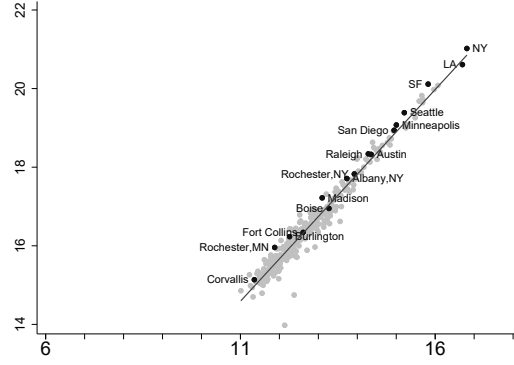
(b) 1950



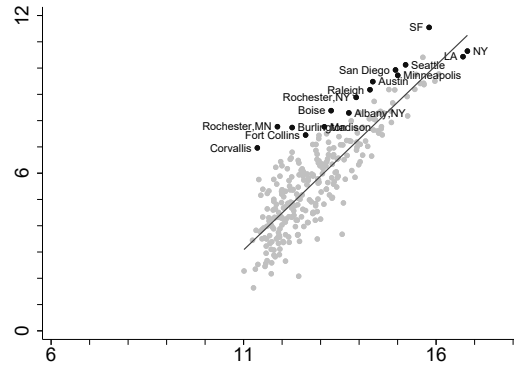
(c) 1900

Note: (a) Share of population, patents and total output by deciles of city size and rural status for 2010. (b) Share of population and patents by deciles of city size and rural status for 1950. (c) Same as (b) but for 1900. In each panel, black bar is population share and white bar is patent share. In panel (a) the gray bar is output share.

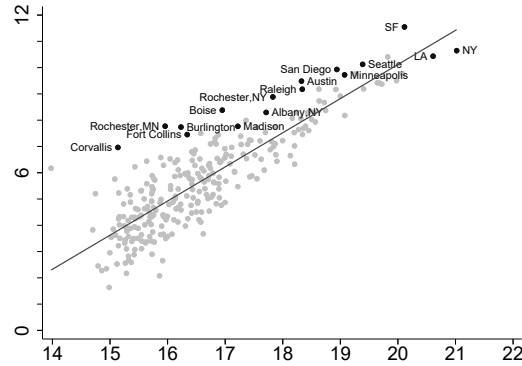
Figure 2: Joint distribution of output, patents and city size in 2010.



(a) Output vs Pop. 2010



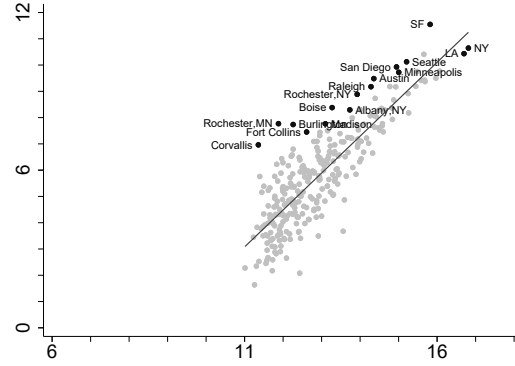
(b) Patents vs Pop. 2010



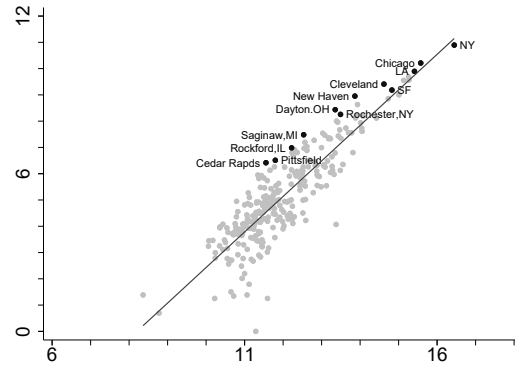
(c) Patents vs Output. 2010

Note: Each panel shows a scatter plot and OLS regression line. (a) $\ln(\text{Output})$ vs. $\ln(\text{Population})$ 2010; $\beta = 1.08, s.e. = 0.013$. (b) $\ln(\text{Patents})$ vs. $\ln(\text{Population})$ 2010; $\beta = 1.41, s.e. = 0.053$. (c) $\ln(\text{Patents})$ vs. $\ln(\text{Output})$ 2010; $\beta = 1.29, s.e. = 0.047$.

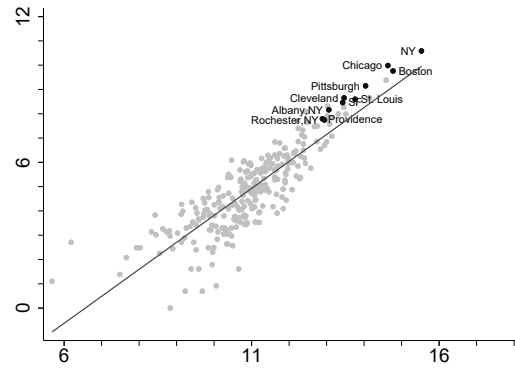
Figure 3: Joint distribution of population and patents over time.



(a) Patents vs Pop. 2010



(b) Patents vs Pop. 1950



(c) Patents vs Pop. 1900

Note: Each panel shows a scatter plot and OLS regression line. (a) $\ln(\text{Patents})$ vs. $\ln(\text{Population})$ 2010; $\beta = 1.41, s.e. = 0.053$. (b) $\ln(\text{Patents})$ vs. $\ln(\text{Population})$ 1950; $\beta = 1.35, s.e. = 0.046$. (c) $\ln(\text{Patents})$ vs. $\ln(\text{Population})$ 1900; $\beta = 1.11, s.e. = 0.041$.

3. Literature

The relationship between how people are distributed across cities and the level and growth of economic output is the subject of a large literature. We here consider only papers most relevant to our inquiry, and refer the reader to Fajgelbaum and Gaubert (2020) and Rossi-Hansberg et al. (2026) for more exhaustive surveys. We organize our discussion of the literature around the four mechanisms discussed above.

A Urban Scale Effects on Output

Ciccone and Hall (1996) is an early effort to estimate urban scale effects, explicitly accounting for differences in both physical and human capital. Their benchmark estimate implies an elasticity of city-level productivity with respect to city population of 3.4%. The estimate of this parameter is indirect.⁶

Most other efforts to estimate urban scale effects rely on regressions of the logarithm of wages or output on city size. Glaeser and Gottlieb (2009) estimate the relationship between city level output and population using data on US cities in 2000, finding that city size elasticity of output is about 13%. Glaeser and Gottlieb (2008) look at the relationship between wages and city size using a large cross-section of US workers. They estimate that the city size elasticity of wages is about 4.1%. The estimates in these two papers do not correct for a number of possible confounders, including the propensity of more productive people to sort into cities, the possibility that people accumulate unobserved human capital more quickly in cities, and the possibility that people accumulate at places that are intrinsically more productive. Subsequent literature has proposed a variety of solutions to these problems (see Rosenthal and Strange (2004) and Combes and Gobillon (2015) for surveys).⁷

Each of Combes et al. (2008), De la Roca and Puga (2017) and Duranton and Puga (2023) relies on a panel of individual workers to examine the relationship between wages and city size. In French data, Combes et al. (2008) find that the city size elasticity of wages is about 5.1%. After controlling for individual fixed effects, this drops to about 3.7%. Using Spanish data, De la Roca and Puga (2017) conduct a similar exercise and find that the city size elasticity of wages is about 5.1%. After controlling for individual fixed effects, this drops to about 2.2%. Duranton and Puga (2023) replicate De la Roca and

⁶Ciccone and Hall rely on state level output data and county level population, and infer county level returns to scale from the relationship between state output and county population heterogeneity. With increasing returns to scale in county population, in two otherwise similar states, a state with more county level population heterogeneity has higher output.

⁷We note that the empirical literature is generally careful to distinguish between the effects of city size and city density. To simplify our analysis, as is common in much of the theoretical literature, we abstract from this distinction and treat the two concepts as interchangeable.

Puga (2017) for the panel of US workers described by the NLSY79 and find that the city size elasticity of wages is about 7.6%. This drops to 4.4% after controlling for individual fixed effects.

B Urban Scale Effects on Human Capital

Glaeser and Maré (2001) advance the argument that workers in cities produce more output not because the city makes their existing human capital more productive, but rather because being in the city raises their human capital.

Three of the papers discussed in the previous section use panel data with worker fixed effects to examine the effect of city size on wages, Combes et al. (2008), De la Roca and Puga (2017) and Duranton and Puga (2023). Each finds that including worker fixed effects reduces the effect of city size on wages. This is consistent with more productive people sorting into more productive places. It is also consistent with people being a priori identical, but accumulating unobserved human capital more rapidly in bigger cities. In the first case, people are more productive when they arrive in bigger cities, in the second, they become more productive after they arrive.

De la Roca and Puga (2017) and Duranton and Puga (2023) estimate regression specifications that allow us to distinguish these two hypotheses and find that more rapid unobserved human capital accumulation in larger cities can account for most or all of the ‘sorting’ effect estimated in Combes et al. (2008). In particular, the estimated elasticities of unobserved human capital with respect to city size are 2.9% in De la Roca and Puga (2017) and 3.1% in Duranton and Puga (2023). On the other hand, using large administrative data sets for the US, Card et al. (2024) also investigate the effect of city size on human capital accumulation, and find that the effect is not economically important. Because of these conflicting results, the importance of city size for human capital accumulation remains somewhat uncertain.

The above papers all consider the accumulation of human capital associated with cities as an essentially static effect. By contrast, another strain of the literature builds mechanisms by which human capital accumulation in cities can be a driver of long run growth. In the model of Black and Henderson (1999), educational spending is linearly converted into human capital; when the productive externality from the average level of human capital in a city is sufficiently large, educational investment alone is sufficient to produce endogenous growth. Similarly in Crews (2026) there is a local externality in the process of human capital accumulation. Along with constant returns to scale in the accumulation of human capital, this is sufficient to produce long run growth. In Duranton and Puga (2023), there are two drivers of long-run growth. The first, is exogenous technological progress, which takes the form of non-negative stochastic

shocks to the production amenities of each location. The second is growth of human capital which results from two spillovers: from the previous generation and from fellow workers in the city where an individual lives.⁸

In none of the papers just mentioned does R&D or innovation take place. This is in contrast to the approach in most growth literature, in which technological progress, reflected in rising TFP and resulting from explicit research and development effort, is generally viewed as growth's most important driver (Jones (2002)). The next section discusses literature in which urban structure is linked to long-run growth through this channel.

C Urban Scale Effects on Innovation and the Link from Innovation to TFP Growth

A large literature establishes that, for people working in knowledge intensive activities, proximity to other people working in similar industries has important effects on productivity, and also that the benefits of proximity fall off rapidly with distance. For example, Arzaghi and Henderson (2008) show that a few hundred meters of distance from an incumbent firm has a large impact in the location choice of an entrant, while Atkin et al. (2022) shows the importance of face-to-face contact for patent citations. Similarly, Carlino and Kerr (2015) use results in Rosenthal and Strange (2003) to calculate that the benefits of proximity decrease about five times more quickly with distance for software production than for metal fabrication. There is also evidence that inventive or innovative activity is much more likely to cluster together than it would if firms chose locations at random, e.g. Inoue et al. (2019). Desmet and Rossi-Hansberg (2014) develop a spatial model that provides micro-foundations for the relationship between density and innovation. For a useful survey of both literatures, see Carlino and Kerr (2015) and Kerr and Kominers (2015). These papers strongly suggest the existence of scale effects that are more important for innovation and invention than for most other types of economic activity, but they are less helpful for thinking about how scale effects vary with the size of a city.

Carlino et al. (2007) applies more directly to our case. This paper estimates a cross-sectional regression of patents per person on employment density in US MSAs around 2000. They estimate an elasticity of 17-20%. Finally, Moretti (2021), constructs a panel of US inventors and their patenting activity by year, sector, and BEA economic area (slightly larger than an MSA). Controlling for inventor fixed effects, this paper estimates

⁸In Davis et al. (2014) there are agglomeration effects from urban density to TFP that amplify the effect of growth in the technologies related to production of output, business investment, and infrastructure investment. Calibrating their model to US data, they find that these agglomeration economies increase the growth rate of output by about 10%. But these underlying rates of technological progress are exogenous to the model.

that the elasticity of inventions per inventor with respect to the number of inventors in a sector-cluster is 5% to 9%, depending on specification, with a preferred estimate around 6%.⁹¹⁰

Using estimates of the relationship between city size and innovation to understand how changes in city sizes affect aggregate output is complicated for two reasons. First, innovation that takes place in a given city not only raises output in that city, but in the country as a whole. Over the long time periods that we consider, we ignore issues related to the rate at which technology diffuses across space. Second, while estimates of the empirical relationship between city size and the production of innovation are described by simple log linear relationships, the relationship between innovation and growth in TFP is more complicated. Jones (2002) and Bloom et al. (2020), provide extensive evidence that the amount of TFP growth resulting from a given quantity of R&D is a negative function of the level of TFP. This “fishing out effect” results from ideas being harder to find as more of them have been discovered.

Using the preferred parameterization from Bloom et al. (2020), a doubling of the level of TFP reduces by about 7/8 the amount of TFP growth that results from a given amount of R&D effort. As we show below, this implies that the reduction in the amount of effective R&D that would result if city sizes had been smaller in the past would produce only a modest decline in the level of TFP today.

D *Improved Sorting*

A large literature focuses on differences in worker productivity across cities, the potential for redistribution of workers toward high-productivity cities to raise aggregate output, and the barriers impeding such a redistribution.

Hsieh and Moretti (2019) find that, after correcting for cross-MSA differences in education, log wages in San Francisco, San Jose, and New York were about 0.465 above the US mean in 2009. They attribute these wage differences to differences in the marginal product of labor, suggesting large potential efficiency gains from moving people to high-productivity cities. In their main counterfactual case, they restrict cross MSA variation in housing prices in 2009 to its 1964 levels. This leads to an equilibrium migration of population towards higher wage cities. For example, New York, San Francisco, Flint, and San Jose all see their counterfactual employment growth rates more than double. This

⁹The fact that patenting seems to increase more rapidly with city size than does output is also visible comparing the first two panels of figure 2.

¹⁰We note that Wiebe (2026) conducts a reanalysis of Moretti (2021) and concludes, after correcting errors in Moretti’s analysis, that the effect of city size on patenting is close to zero in some (but not all) of Moretti’s regression specifications. We treat Moretti’s estimates as a lower bound on the effect of city size on patenting. Wiebe’s results suggest an even smaller lower bound.

migration increases the level of 2009 output by the same amount as an increase in the 1964-2009 growth rate from its baseline of 0.80% to 1.49% per year. Over the course of their 1964-2009 study period, this results in 8.9% higher counterfactual output in 2009.

Fajgelbaum and Gaubert (2020) consider a similar problem. They describe the effect of reallocating people on welfare, average income adjusted for local prices, rather than output, as in Hsieh and Moretti (2019). Importantly, they also allow for the possibility of transfers across cities. Calibrating this model to fit US data for around 2000, they find that the optimal reallocation of people across cities can increase welfare by between 2 and 6%. Rossi-Hansberg et al. (2026) generalizes Fajgelbaum and Gaubert (2020) to allow for multiple occupations and costly trade in goods. Calibrating their model to US data for around 2010, they conclude that the optimal reallocation of population from the 2010 level would increase welfare by less than 1%.¹¹ Duranton and Puga (2023) develop a more elaborate model with many of the same underlying mechanisms. Growth is driven by exogenous growth in the overall population, mediated by an endogenous assignment of people to cities. They conduct a counterfactual in which barriers to migration into the largest cities are reduced. They find that moving 11% of the population gives an 8% increase in the level of output.

In all of these papers, differences in city-level productivity are due to one or both of the following: first, agglomeration effects like the ones we discussed above, and second, differences in location specific productivity that are not a function of city size. Hsieh and Moretti fix city specific productivity exogenously; it does not vary with the size or demographic composition of the city. By contrast, in Fajgelbaum and Gaubert (2020) city specific productivity depends on the number of educated and uneducated workers in addition to an exogenously fixed component. In Rossi-Hansberg et al. (2026) there is city-specific productivity that varies across intermediate goods sectors as well as a city-wide productivity effect that depends on occupation shares and total employment. The authors are able to invert the model to back out city-sector productivity levels that are combinations of these endogenous and exogenous factors, with e.g. New York having the highest productivity in finance and insurance and San Jose having the highest productivity in computers and electronics. In Duranton and Puga (2023) there is city-specific productivity that evolves exogenously and a positive agglomeration effect from the (endogenous) quantity of human capital. In the model of Crews (2026)

¹¹One reason that Rossi-Hansberg et al. (2026) and Fajgelbaum and Gaubert (2020) conclude that there is less gain to be had from shifting people across locations than Hsieh and Moretti may be that people are better sorted in the later period that Fajgelbaum and Gaubert (2020) and Rossi-Hansberg et al. (2026) consider than the earlier benchmark year for the Hsieh and Moretti calculation. The data weigh against this. Hsieh and Moretti show that there was less cross-city wage dispersion in the earlier period, and hence, probably less potential benefit from shifting people across cities. Second, the Fajgelbaum and Gaubert (2020) and Rossi-Hansberg et al. (2026) finding that city sizes have only a small effect on the level of output is broadly consistent with our findings below.

there is also a term for local fundamental productivity; he takes this from the city fixed effects estimated in Duranton and Puga (2023), which vary by a factor of roughly 1.25 in comparing the highest and lowest values among MSAs.

Of the two components of city-level productivity differences just described, one, the agglomeration effect, is at the core of the quantitative analysis that we present below. The question is how to think about the other component: exogenous city-specific productivity differences. In principle, in a world where variation in this sort of exogenous productivity was large, changing the maximum size of cities would affect aggregate productivity by affecting the number of people who could live in the best places. If the population of the San Jose MSA was limited to one million, half of the current residents would have to live somewhere with lower exogenous productivity. From this perspective large cities are important not because of largeness *per se*, but because there is a shortage of locations with high exogenous productivity, and large cities allow more people to live in those special locations.

In the rest of the paper, we rule out this sort of exogenous variation in location specific productivity. While it is a useful modeling tool, we do not see it as something that is meant to be an actual description of reality. An additional problem with incorporating this feature into our analysis is that it would require our being able to specify the distribution of productivity in places that currently do not have cities in them, which is a challenge the literature has not taken on.

None of this is to say that barriers to mobility of the type studied by e.g. Hsieh and Moretti (2019) are not important. Even if there are no differences in fundamental productivity across locations, agglomeration effects on their own are enough to establish that the observed distribution of population across cities might not be efficient. In this paper we put that issue aside, however, and focus on the more narrow question of whether changes in the maximum size of cities on their own would have a large effect on output.

Summing up, Hsieh and Moretti (2019), Fajgelbaum and Gaubert (2020), and Rossi-Hansberg et al. (2026) consider cities for which city level productivity consists of a fixed exogenous component and possibly an agglomeration effect relating city size and composition to output. While Hsieh and Moretti (2019) indicates large gains from shifting population to reduce cross-city wage gaps, later analyses that allow for agglomeration economies, transfers, and cross city trade, revise these estimates down considerably. None of the three papers considers the role of agglomeration economies in technological progress, either in the large cities where most new technologies originate or in the rest of the country, to which they diffuse.

4. Urban Scale Economies for Output and Human Capital

We would like to calculate the total value of output for a counterfactual version of the US in which certain cities are constrained to be smaller than they are. We treat MSAs as the real world analog of our theoretical cities, and index them by $i = 1, \dots, n$. To represent the non-MSA sector of the economy, we use the index $i = 0$. (We often abuse language and refer to i as indexing “cities.”) Y_{it} denotes output of location i in decade t , L_{it} population, K_{it} physical capital, ℓ_{it}^Y the fraction of the population engaged in the production of output, h_{it}^Y the human capital of workers engaged in producing output, and A_{it} is city-level productivity in producing output. We assume that a city transforms inputs into outputs according to

$$Y_{it} = A_{it} (K_{it})^\gamma \left(h_{it}^Y \ell_{it}^Y L_{it} \right)^{1-\gamma}. \quad (1)$$

We are interested in understanding how changing the size distribution of cities would affect aggregate output. To proceed, we decompose A_{it} into three components. The first term is a time-specific national component of productivity common to all cities, $\bar{A}_t$. Growth in $\bar{A}_t$ is growth in TFP, a quantity that is central to our analysis. $\tilde{A}_{it}$ is the urban scale effect for output, which we discuss below. Finally, $\hat{A}_{it}$ is a city-decade specific idiosyncratic term. This term reflects transitory idiosyncratic shocks to a location’s productivity. We assume that $\sum_i \hat{A}_{it} = n + 1$, so that its mean value is one in each period. Altogether, we have,

$$A_{it} = \hat{A}_{it} \bar{A}_t \tilde{A}_{it}. \quad (2)$$

We assume that the urban scale effect for output depends on city population. Because the rural sector is large and unitary, we must find some other way to assess scale effects for the rural area. To proceed, we note that the minimum population for an MSA is 50,000, and assign rural areas the same scale effect as a city of this size, regardless of how many people live in rural areas. To implement this, we define $\tilde{L}$ as the population that is relevant for urban scale effects on output.

$$\tilde{L}_{it} = \begin{cases} L_{it} & \text{for } i > 0 \text{ (MSAs)} \\ \underline{L} & \text{for } i = 0 \text{ (rural areas).} \end{cases} \quad (3)$$

For MSAs, $\tilde{L}$ is the same as population, but for the aggregate rural area we cap urban scale effects at the level of the smallest MSA, that is, at $\underline{L} = 50,000$. As a practical matter, the size of the rural population is invariant across all of the counterfactuals we conduct in the body of the paper, and so this assumption will not affect our main results. This assumption is required in Appendix A, however, and introducing it here allows consistent notation throughout.

Using this notation, we specify the urban scale effect for output as

$$\tilde{A}_{it} = \tilde{L}_{it}^{\sigma_A} \quad (4)$$

We discuss the values we use for the parameter σ_A later in this section. Combining equations (1), (2), and (4) results in a production technology that nests production functions commonly used to study systems of cities, e.g., Desmet and Rossi-Hansberg (2013), Duranton and Puga (2023).

We assume that physical capital is freely mobile among cities and that this equalizes its marginal product. This implies that,

$$\frac{Y_{it}}{K_{it}} = \frac{Y_t}{K_t} \quad \forall i \quad (5)$$

Substituting (5) into the production function (1) and rearranging, we have

$$Y_{it} = A_{it}^{1/(1-\gamma)} \left(\frac{K_t}{Y_t} \right)^{\gamma/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it} \quad (6)$$

Summing over all cities, we get aggregate output,

$$Y_t = \left(\frac{K_t}{Y_t} \right)^{\gamma/(1-\gamma)} \sum_i A_{it}^{1/(1-\gamma)} h_{it}^Y \ell_{it}^Y L_{it}. \quad (7)$$

We would like to compare the observed, or “base” case, to an alternative where some cities take counterfactual sizes. When necessary, we indicate the value of variable X in the two cases with superscripts, X^{base} and X^{alt} .

We assume that the aggregate ratio of capital to output, K/Y , does not differ between the two cases at a point in time. The simplest justification for holding the K/Y ratio fixed is if the country is open to the world capital market with the interest rate not varying across scenarios. Alternatively, Romer (2012) shows that if capital accumulates via a fixed investment rate (as in Solow (1957)) then the capital to output ratio is constant along any balanced growth path.¹² Empirically, Feenstra et al. (2015) show that across countries capital-output ratios do not vary systematically with income. Similarly Jones and Vollrath (2020) show the relative constancy of this ratio over time within countries that are arguably close to their balanced growth paths.

Assuming a fixed K/Y ratio allows us to incorporate “induced capital accumulation” (Klenow and Rodriguez-Clare, 1997, Hall and Jones, 1999): we expect that a change in

¹²If differences between our scenarios were purely in terms of the level of productivity but not its growth rate, this condition would hold. However, in later sections, we will allow for reduced agglomeration to affect the growth rate of aggregate productivity, and so the condition is no longer exact. If limiting agglomeration slowed technological progress, then the K/Y ratio would rise, partially offsetting the effect of slower productivity growth on the level of output. Thus differences in output between the baseline and alternative cases, which we find to be small under our assumption of fixed K/Y , would be even smaller if we did not make that assumption.

city level productivity to affect the level of income, and hence the quantity of investment. This change in investment then feeds back to affect output.

We restrict attention to alternative cases where a city's population is unchanged but the size of the urban scale effect on productivity ($\tilde{A}_{it}$) is reduced to that of a smaller city. All other characteristics of the city, h_{it} , ℓ_{it}^Y , and the city-decade component of productivity, $\hat{A}_{it}$, remain constant. We can also imagine this occurring if the observed population of the city L_{it}^{base} is divided into $\frac{L_{it}^{base}}{L_{max}}$ daughter cities, each with population L_{max} , with human capital equally divided among them, and with all of the daughter cities having the same values of $\hat{A}_{it}$ and ℓ_{it}^Y as the original city. We assume that non-metropolitan output does not change between base and alternative cases.

Because congestion effects are not part of the production process of equation (1), and because production in the absence of the agglomeration effect is CRS, perfect mobility of all factors of production would lead to an equilibrium in which all production took place in the city with the highest value of $\hat{A}_{it}$. We are thus implicitly considering equilibrium population levels that are partly determined by an unspecified congestion process.

With these assumptions in place, we can use equation (7) to compare aggregate output in economies with different values of A_{it} . Multiplying each term in the sum on the right hand side of (7) by $\left(\frac{A_{it}^{base}}{A_{it}^{alt}}\right)^{1/(1-\gamma)}$ and using equation (7) again to simplify, we have

$$Y_t^{alt} = \sum_i Y_{it}^{base} \left(\frac{A_{it}^{alt}}{A_{it}^{base}} \right)^{1/(1-\gamma)}.$$

Dividing by Y_t^{base} gives

$$\frac{Y_t^{alt}}{Y_t^{base}} = \sum_i \frac{Y_{it}^{base}}{Y_t^{base}} \left(\frac{A_{it}^{alt}}{A_{it}^{base}} \right)^{1/(1-\gamma)}. \quad (8)$$

We would like to evaluate the effect on the output of a particular city of constraining its productivity to that of a city of size no greater than L_{max} . Because city size enters a city's TFP, A_{it} , only through the static scale effect of equation (4), the ratio of observed to counterfactual city productivity is,

$$\frac{A_{it}^{alt}}{A_{it}^{base}} = \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\sigma_A} \right). \quad (9)$$

Using equation (9) and (8) together, we can evaluate aggregate output for a counterfactual system of cities in which all cities with population above the threshold level L_{max} have their productivity reduced to that of a city of the threshold size. The resulting change in aggregate output is,

$$\frac{Y_t^{alt}}{Y_t^{base}} = \sum_i \frac{Y_{it}^{base}}{Y_t^{base}} \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\sigma_A/(1-\gamma)} \right) \quad (10)$$

In equation (10) we see the advantage of restricting attention to our particular counterfactuals. For these counterfactuals we can evaluate the change in aggregate output without measures of city-specific physical and human capital or of the other parts of city productivity term, $\hat{A}_{it}$ and $\bar{A}_t$. We require only city population and output.¹³

As in Fajgelbaum and Gaubert (2020), Rossi-Hansberg et al. (2026), and Hsieh and Moretti (2019), equation (10) evaluates the change in output under counterfactual distributions of people across cities. However, the foundations of our approach are different in several ways. First, the papers just mentioned each describe a complete general equilibrium and rely on different descriptions of the production process. Indeed, both Fajgelbaum and Gaubert and Rossi-Hansberg et al. are based on a more detailed description of agglomeration externalities than the one we use. Second, all three restrict attention to equilibrium comparative statics. Third, all assume that worker heterogeneity is completely described by their respective models. Finally, all three papers rest on the estimation of many structural parameters.

In contrast, we do not specify the complete general equilibrium. We consider out of equilibrium comparative statics, and we consider a simple description of the production process. Our approach has two advantages. First, it only requires estimates of the static agglomeration effect σ_A and labor share γ , both intensively researched parameters. Second, our comparative static is robust to arbitrary descriptions of worker heterogeneity and how workers sort across cities. One can imagine deriving the analog of equation (10) in the context of any of the models developed in Fajgelbaum and Gaubert (2020), Rossi-Hansberg et al. (2026) or Hsieh and Moretti (2019), and leaving the rest of our analysis untouched. The results presented in these papers lead us to suspect that this would not qualitatively change our conclusions.

A Human Capital Extension

We now turn to our description of urban scale economies for unobserved human capital. To begin, we assume that human capital is a function of city-decade specific inputs (years of education and their quality), which we denote S_{it} . We further allow the Mincerian return to these inputs, denoted ϕ_{it} to vary at the city-decade level. Finally, we allow an urban scale effect similar to the one for producing output, represented by the parameter σ_H :

$$h_{it}^Y = \exp(\phi_{it} S_{it}) \tilde{L}_{it}^{\sigma_H}, \quad (11)$$

¹³Estimates of city-specific productivity, $\hat{A}_{it}$, for example, face a series of econometric problems. Does a particular city produce high output relative to its human capital because it has a high idiosyncratic productivity due to location or institutions, or because we do not properly measure the quality of human capital? This problem will recur in our analysis of city level research productivity. These problems are carefully described in Combes et al. (2010) and Glaeser and Gottlieb (2008).

Next, we use equation (7) to write aggregate output for a counterfactual case where city size is capped at L_{max} and multiply the right hand side by

$$\left(\frac{A_{it}^{base}}{A_{it}^{base}} \right)^{1/(1-\gamma)} \frac{h_{it}^{Y,base}}{h_{it}^{Y,base}}. \quad (12)$$

Following the same logic that leads from equation (7) to (10), we arrive at the corresponding expression for aggregate output when human capital production is subject to scale effects,

$$\frac{Y_t^{alt}}{Y_t^{base}} = \sum_i \frac{Y_{it}^{base}}{Y_t^{base}} \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\frac{\sigma_A}{1-\gamma} + \sigma_H} \right) \quad (13)$$

Comparing equation (13) to equation (10), we see that the two expressions are identical save for the interpretation and magnitude of the exponent on the term $\left(\frac{L_{max}}{\tilde{L}_{it}} \right)$. Therefore, for the purpose of evaluating counterfactual scenarios, we evaluate the model with or without scale effects in the production of human capital by varying the magnitude and interpretation of this exponent.¹⁴

B Parameterization

Given the data already in hand, evaluating equation (13) requires only that we evaluate $\frac{\sigma_A}{1-\gamma} + \sigma_H$. To evaluate σ_A and σ_H , we rely on the literature estimating the relationship between city size and productivity that was discussed in Section 3

Table 1 lists estimates of these parameters derived from six papers described in section 3. With the exception of Ciccone and Hall (1996), these papers rely on regressions of log wages on log city sizes in cross section or panel data. To convert estimates of the elasticity of wages with respect to city size into our desired parameter σ_A , note that from (8) we can derive the relationship between city size and output as,

$$\frac{\partial \log Y_{it}}{\partial \log L_{it}} = 1 + \frac{\sigma_A}{1-\gamma}. \quad (14)$$

If we make the further assumption that wages reflect the marginal productivity of labor, then we can also derive the relationship between city size and wages,

$$\frac{\partial \log w_{it}}{\partial \log L_{it}} = \frac{\sigma_A}{1-\gamma}. \quad (15)$$

¹⁴Our description of the effect of city size on human capital simplifies the problem by assuming that human capital accumulated in a city can never migrate to another city. A better, but much less tractable, description of the process would allow people to accumulate human capital in one city and employ it in another. Implementing such a model would require vastly more data than the exercise we conduct. A tractable alternative to our approach would be to allow human capital to be freely mobile across cities in parallel to physical capital. However, in this case, rather than being too attached to the cities where it is accumulated, human capital might be too unattached to the people who accumulate it.

Table 1: Estimates of σ_A

σ_A	σ_H	$\frac{\sigma_A}{1-\gamma} + \sigma_H$	Source	Data
3.4%	~ 0	5.1%	Ciccone and Hall (1996)	US, State output, 1988
2.5%	$\equiv 0$	3.7%	Combes et al. (2008)	French, Ind. wages, 1976-98
1.5%	2.9%	5.1%	De la Roca and Puga (2017)	Spanish, Ind. wages, 2004-9
2.9%	3.1%	7.6%	Duranton and Puga (2023)	US, Ind. wages, ca. 1979-2020
2.7%	$\equiv 0$	4.1%	Glaeser and Gottlieb (2008)	US, Ind. wages, 2000
8.6%	$\equiv 0$	13%	Glaeser and Gottlieb (2009)	US, MSA output, 2000

Note: Various estimates of the static scale effect, σ_A and the human capital scale effect, σ_H from the literature. “ $\equiv 0$ ” indicates a quantity implicitly assumed to be zero.

We follow the standard in the growth literature and set the capital share of output, γ , to 0.33.

The only paper in Table 1 that does not take this direct regression approach is Ciccone and Hall (1996). Although their estimation strategy is indirect, the parameters that they estimate correspond closely to ours.¹⁵

The estimates of $\frac{\sigma_A}{1-\gamma} + \sigma_H$ presented in table 1 range from about just under 4% to 13%. However, four of the six estimates are within about 1% of the bottom end of this range. Given this, our preferred value of $\frac{\sigma_A}{1-\gamma} + \sigma_H$ for our calculations is 4%. With that said, given that there is still some variation around this estimate, we also consider 8% and 12% in our calculations.

To evaluate equation (13) we use the data on city level output and population described above. We evaluate the static effect of agglomeration for data from the year 2010, considering two possible values of maximum city size, $L_{\max}$: one million and 100,000. Note that even our mildest comparative static, capping city size at one million involves a catastrophic reorganization of the economy. The smallest US city with a population above one million in 2010 was Fresno, CA, the 52nd largest MSA in the country. In 2010 the largest 52 cities housed 58% of the population and produced 66% of output. A cap of 100,000 requires reorganizing 261 MSAs.

The results of this exercise are presented in Table 2. For a city size cap of one million, using our preferred value of $\frac{\sigma_A}{1-\gamma} + \sigma_H = 4\%$, the reduction in output is 6%, which suggests that static agglomeration effects are not very important. Even using our high-end value of the exponent, restricting city sizes to one million would reduce output by

Table 2: Output in 2010 for two counterfactual size caps and values of σ_A .

$\frac{\sigma_A}{1-\gamma} + \sigma_H$	$L_{max} = 1m$	$L_{max} = 100k$
0.04	0.94	0.84
0.08	0.88	0.72
0.12	0.83	0.62

Note: Each cell reports output relative to the baseline for a particular cap on city size and value of the exponent in equation (13). For the purpose of this calculation, the rural population is treated as an extra MSA whose output is constant across scenarios and the capital share of output, γ , is equal to 0.33.

only 17%.

5. Urban Scale Economies for Patents

The analysis in Section 4 takes the city invariant component of productivity, $\bar{A}_t$, as given. Changes in this parameter over time reflect technological progress, which we now examine. We proceed in three steps. First, we consider the relationship between city size and the production of patents. Second, we consider the relationship between patents and effective research effort, a term that we define more precisely below. Finally, we consider the relationship between effective research effort and changes in $\bar{A}_t$. The first of these three steps is taken in this section, while the second and third are in Section 6. Combining the first two steps we can investigate the relationship between city sizes and effective research effort, decade by decade. Putting the three steps together, we describe the relationship between the distribution of city sizes and the speed of technological progress.

Our analysis of the relationship between city sizes and patents parallels our analysis of the relationship between city sizes and output. The number of patents produced in a city-decade, P_{it} , depends on the size of the research labor force (specifically, city population multiplied by the share of people working in R&D, ℓ_{it}^R), the human capital of those research workers, h_{it}^R , and a city-decade patent productivity multiplier, B_{it} , according to the function,¹⁶

$$P_{it} = B_{it} h_{it}^R \ell_{it}^R L_{it}. \quad (16)$$

¹⁵Using their benchmark estimate of $\hat{\theta} = 1.052$ (from Table 1) in their equation (20) and our assumption that the capital share is 0.33, we have that (their notation) $\gamma = 1.034$. Inspection of their equation (3) confirms that this parameter corresponds to our σ_A . Similarly, the benchmark estimate of η is not distinguishable from zero, and inspection of their equation (3) confirms that this parameter corresponds to our σ_H .

¹⁶To simplify the model, we assume that physical capital is not used for the production of research output.

Summing over cities within a year, we have aggregate patent production,

$$P_t = \sum_i B_{it} h_{it}^R \ell_{it}^R L_{it}. \quad (17)$$

City-decade patent productivity, B_{it} , can be decomposed into three components: a time specific national component common to all cities, $\bar{B}_t$; a city specific agglomeration effect that depends on population, $\tilde{B}_{it}$; and, a city-decade specific idiosyncratic term, $\hat{B}_{it}$. More formally,

$$B_{it} = \hat{B}_{it} \bar{B}_t \tilde{B}_{it}. \quad (18)$$

We model the urban scale effect for innovation in the same way we did for output,

$$\tilde{B}_{it} = \tilde{L}_{it}^{\sigma_B}, \quad (19)$$

but with a different value of returns to scale parameter.

We do not restrict the relationship between city-specific output productivity (the $\hat{A}_{it}$'s) and city-specific patent productivity (the $\hat{B}_{it}$'s). Places can be good at one but not the other. Nor do we restrict the relationship between the quality of human capital used to produce output, h_{it}^Y and that used to produce patents, h_{it}^R .

As in the previous section, we consider the thought experiment where the urban scale effect on research productivity, $\tilde{B}_{it}$, takes the value that would hold if the city were constrained to maximum size L_{max} . To evaluate the resulting change in counterfactual patent production, we use the same approach as in our analysis of counterfactual output, adjusting for the absence of capital. To proceed, first use (17) to write aggregate research output for the counterfactual case. Second, multiply the right hand side by $\frac{B_{it}^{base}}{\tilde{B}_{it}^{base}}$. Third, rearrange and use equations (18) and (19) to get

$$P_t^{alt} = \sum_i P_{it}^{base} \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\sigma_B} \right). \quad (20)$$

Finally, divide both sides by P_t^{base} to get

$$\frac{P_t^{alt}}{P_t^{base}} = \sum_i \frac{P_{it}^{base}}{P_t^{base}} \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\sigma_B} \right). \quad (21)$$

In words, equation (21) allows us to calculate the level of patenting under a counterfactual system of cities from observed city and national patenting, observed city populations, and assumed counterfactual city populations.

In fact, equation (21) describes the largest possible change that could result from a change to city sizes. To see this, consider the case in which there is a city of two million people, of whom 20,000 are engaged in research. It seems likely (see next section) that agglomeration effects in research depend on the number of other researchers in a city,

rather than the number of people overall. This means that one could imagine splitting the parent city into two daughter cities, each with one million people, but with one daughter city containing all 20,000 researchers. In that case, patent production would not fall at all. By dividing up the resources devoted to research proportionally with population, as in equation (21), we maximize the effect of reductions in city size on the production of patents.

A Parameterization

The key parameter required for the calculation in equation (21) is σ_B , the effect of city size on patent production. Moretti (2021) is the only estimate based on disaggregated panel data, and suggests values of $\sigma_B \in [0.05, 0.09]$. Carlino et al. (2007) uses only cross-sectional data, and so is less able to address reverse causation and sorting than Moretti (2021), but suggests $\sigma_B \in [0.17, 0.20]$.

For our baseline calculation, we rely on the Moretti (2021) estimate of $\sigma_B = 6\%$ because it is based on higher quality data and an econometric strategy that is better able to control for unobservable attributes of people and cities. With this said, consistent with the estimates from Carlino et al. (2007), we also consider the much larger value $\sigma_B = 20\%$.

Table 3 shows national patent production for 2000-2009 in the alternative case where cities are limited in size relative to the observed case. We consider two values of L_{max} , one million and 100,000, as in table 2. When scale economies in patenting are set at our base-case value of $\sigma_B = .06$ and we cap city size at one million, patent output falls by only 7% relative to baseline. This magnitude is similar to the static urban scale effect on output production shown in table 2 for a similar value of σ_A . We find this surprising. Because patenting is more concentrated in larger cities than output, we expected that restricting city size would be more harmful to patenting than to output. In fact, this intuition appears not to be economically important. Larger declines in aggregate patenting are possible, but require the catastrophic counterfactual changes associated with L_{max} equal to 100,000, or the Carlino et al. (2007) value of σ_B estimated on cross-sectional data rather than the smaller value derived from panel data. In the most extreme case, where we consider the largest plausible value of σ_B and cap city size at 100,000, patenting falls by 42%.

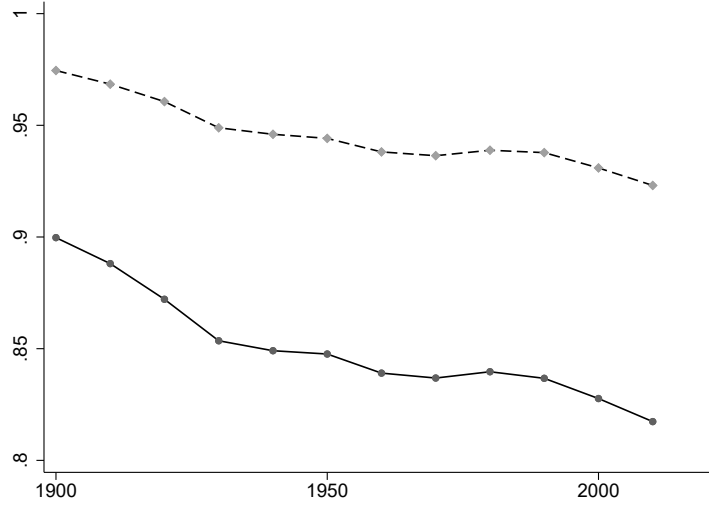
We can also examine the evolution of patenting over time. For this purpose, we restrict attention to our baseline value of $\sigma_B = 0.06$ and calculate how patenting changes in each of our two counterfactual systems of cities. Figure 4 presents our results. As one would expect, restricting city size to a fixed level would have had a much smaller effect on patenting early in the 20th century, when cities were much smaller.

Table 3: Patents during 2000-9 for two counterfactual size caps and values of σ_B .

σ_B	$L_{max} = 1m$	$L_{max} = 100k$
0.06	0.93	0.83
0.20	0.82	0.58

Note: Each cell reports the share of total patents during 2000-2009 relative to totals reported in the CUSP data (Berkes, 2018), for a particular cap on city size and value of σ_B . For the purpose of this calculation, the rural population is treated as an extra MSA whose patents are constant across scenarios.

Figure 4: Share of total patents produced under two hypothetical city networks



Note: Counterfactual patents as a fraction of actual patents reported in CUSP when city sizes are capped at 1m (diamonds) and 100k (squares). Calculations assume $\sigma_B = 0.06$.

6. From Patents to the Level of Technology

The previous section analyzes how patenting in every decade would have differed from its observed value in a counterfactual scenario where city sizes were restricted. We now ask how such a restriction would affect the 2010 level of the time-specific national component of productivity, $\bar{A}_{2010}$.

Our starting point is the framework relating research and technological progress laid out in Bloom et al. (2020). The key equation is

$$\frac{\dot{A}(t)}{A(t)} = \alpha S(t)^\lambda A(t)^{-\beta}. \quad (22)$$

Here, $S(t)$ is research effort and $A(t)$ is the level of technology, both in continuous time.

In their application to the aggregate US economy, $S(t)$ is gross domestic investment in intellectual property products from the National Income and Product Accounts deflated, by a measure of the nominal wage for high-skilled workers. The parameter λ captures the “stepping on toes” effect, whereby the rate of technological progress may not scale

linearly with research effort. The parameter β captures the extent to which ideas become harder to find as more of them have been discovered, the “fishing out effect”.

We make two changes to this framework. First, we allow for the input to technological progress to depend not only on research spending, but also on the degree of agglomeration. Specifically, we define R_t as *effective research effort*, which is S_t adjusted for agglomeration effects in the cities where research takes place. The level of R_t will differ between the base and alternative cases because of limitations on city size. The second change we make is to allow for a time-varying term η_t , in the equation that relates effective research effort and technological progress. This change allows us to solve for an implicit level of effective research effort from the observed time series for productivity.

Making these two changes, and switching to discrete time, equation (22) becomes

$$\Delta \bar{A}_t / \bar{A}_t = \alpha \eta_t R_t^\lambda \bar{A}_t^{-\beta}, \quad (23)$$

and manipulating slightly,

$$\bar{A}_t^{\text{base}} = \bar{A}_{t-1}^{\text{base}} + \left[\alpha \eta_{t-1} (R_{t-1}^{\text{base}})^\lambda \right] (\bar{A}_{t-1}^{\text{base}})^{1-\beta} \quad (24)$$

Using this equation, and data on $\bar{A}$ by decade, we solve for a series for $\alpha \eta_t (R_t^{\text{base}})^\lambda$.

To construct the counterfactual series for effective research effort in the case where city sizes were restricted, R_t^{alt} , we use the machinery relating patents to city sizes from Section 5. Specifically, we use the ratio of $P^{\text{alt}} / P^{\text{base}}$ for every decade that was presented in Figure 4. In mapping from this change in patents to a change in effective research effort, we face two immediate obstacles. First, the relationship between research effort and patenting is not constant over time. For example, between 1910 and 1960, the pace of patent filings by US residents was roughly constant at around 30,000 per year, while the effective number of researchers increased by a factor of more than eight (Berkas, 2018, Bloom et al., 2020). Second, the relationship between patents, on the one hand, and technological progress, on the other, is not stable in either levels or growth rates. For example, the number of patents granted in the US increased more than five-fold between 1980 and 2020, having less than doubled in the previous four decades, but there was no corresponding jump in the rate of TFP growth (USPTO, 2024).

To overcome the first of these obstacles, we assume that the amount of effective research effort required to produce a patent is constant only within a decade, that is:

$$P_{it} = \mu_t R_{it}, \quad (25)$$

where μ_t describes the time varying proportionality between effective research effort and patents.

We overcome the second problem with the time-varying term η_t in equation (23). This parameter allows the relationship between effective research effort and technological progress to vary by decade.

Under these two assumptions,

$$\frac{\alpha\eta_t(R_t^{alt})^\lambda}{\alpha\eta_t(R_t^{base})^\lambda} = \left(\frac{P_t^{alt}}{P_t^{base}}\right)^\lambda \quad (26)$$

$$\alpha\eta_t(R_t^{alt})^\lambda = \left(\frac{P_t^{alt}}{P_t^{base}}\right)^\lambda \alpha\eta_t(R_t^{base})^\lambda \quad (27)$$

The resulting values of $\alpha\eta_t(R_t^{alt})^\lambda$ can then be accumulated forward using the difference equation,

$$\bar{A}_t^{alt} = \bar{A}_{t-1}^{alt} + \left[\alpha\eta_{t-1}(R_{t-1}^{alt})^\lambda\right] (\bar{A}_{t-1}^{alt})^{1-\beta}. \quad (28)$$

This yields the desired path of $\bar{A}$ under the alternative case, and in particular its value in 2010, $\bar{A}_{2010}^{alt}$.

Implementing this process requires values for the two key parameters describing the process of technological progress in equation (23). In their baseline case, Bloom et al. (2020) assume $\lambda = 1$, so that the stepping on toes effect does not operate. Under this assumption, they estimate that $\beta = 3.1$. As an alternative they consider $\lambda = 3/4$. In this case, they estimate that $\beta = 2.4$. In the analysis that follows, we use both pairs of parameterizations. In addition, we also consider a “naive” parameterization of $\lambda = 1, \beta = 0$, where both the stepping on toes effect and the negative effect of current technology on the ease of finding new technologies are absent. Our measure of the level of technology in the base case, $\bar{A}^{base}$ is constructed based on estimates of decadal TFP growth from Bloom et al. (2020) and Gordon (2016a).¹⁷

Two issues regarding our calculations require further discussion. The first is that the estimates of TFP growth used by Bloom et al. and Gordon, reflect not only technological progress, which is what $\bar{A}$ is supposed to be, but also the static urban scale effect discussed in Section 4. In the presence of static urban scale effects, the growth of cities will cause a standard measure of TFP to overstate the speed of technological progress. We elaborate in Appendix A. Under mildly restrictive assumptions we construct a series for technological progress that corrects observed TFP for the growth rate of urban scale effects. This correction is small for the period 1900-2010, in the range of one-tenth of a percent per year. We could easily use this adjusted series as our measure of $\bar{A}_t$. However, to simplify our exposition and to maintain comparability with other literature, we instead equate observed TFP growth with technological progress.

¹⁷Specifically the data are taken from the file AggregateBLSIPP.m in the replication package of Bloom et al. (2020). Their data for the period after 1950 comes from the Bureau of Labor Statistics Private Business Sector multifactor productivity growth series, adding back in the contributions from R&D and IPP. Their data for prior to 1950 come from Gordon (2016a)

The second issue is how to think about technological progress in the rest of the world. In practice, ideas cross borders easily, and so if a reduction in city size had led to less new technology being invented in the US, this deficiency would have largely been made up by innovation from abroad. To address this issue we could impose the same restrictions on global city sizes as we impose in the US. Using global data on city populations and research output, we could then perform an analysis of the effect of this size restriction. Unfortunately, data on city research effort at the global level are not available. As the best practical alternative to an analysis of the entire world, we assume that technological progress in the US results entirely from US research effort.

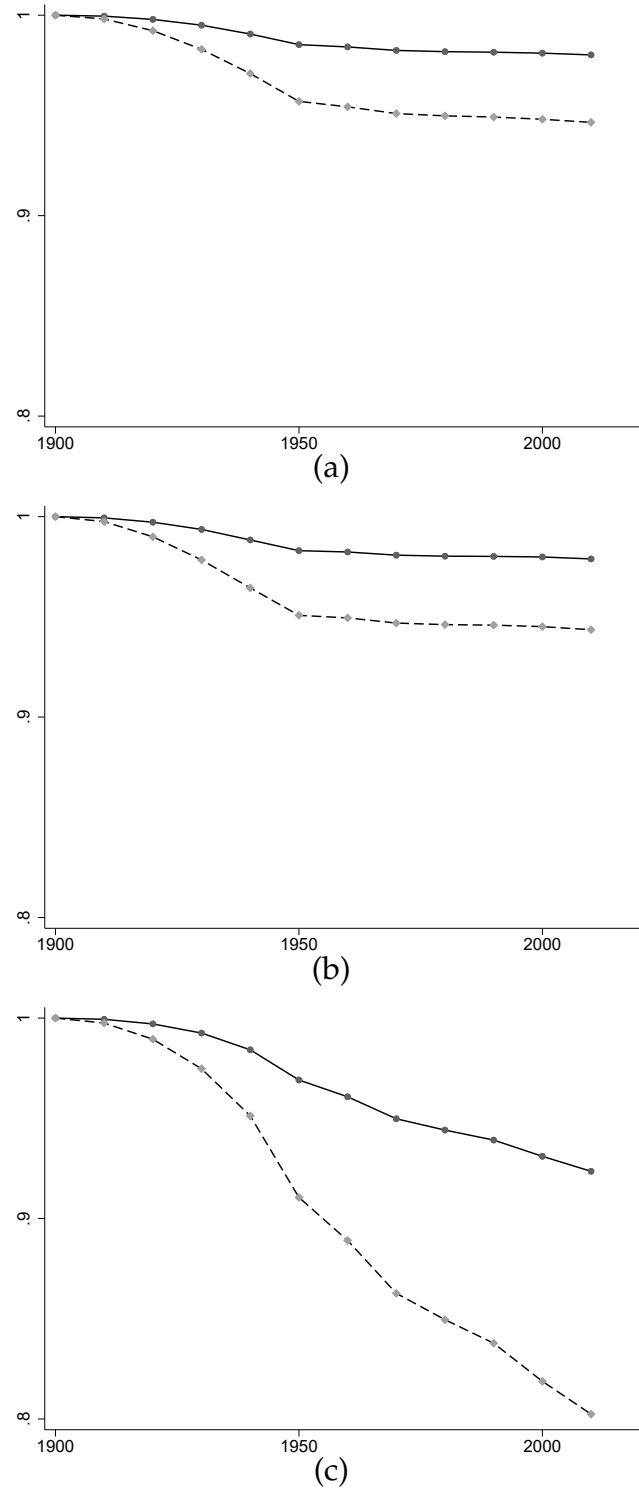
Figure 5 shows our results. Each panel reports the ratio of $\bar{A}_t^{\text{alt}} / \bar{A}_t^{\text{base}}$ in the counterfactual cases where city populations are restricted to one million or 100,000. The three panels differ from each other in parameters that govern the relationship between effective research effort and TFP. Figure 5(a) and (b) use the two sets of parameters considered by Bloom et al. (2020), $(\lambda = 1, \beta = 3.1)$ and $(\lambda = .75, \beta = 2.4)$. Figure 5(c) considers a “naive” parameterization of $\lambda = 1, \beta = 0$, where the stepping on toes effect and the negative effect of current technology on the ease of finding new technologies are both absent.

As the figure shows, the cumulative effect of reduced research output has a small effect on the level of productivity for 2010 under either of the parameterizations used by Bloom et al. (2020). If we restrict attention to the empirically relevant cases where $\lambda = 0.75, \beta = 2.4$ or $\lambda = 1, \beta = 3.1$, then when $\sigma_B = 0.06$ and city size is limited to one million, $\bar{A}$ in 2010 is only about two percent lower than the baseline in both alternative cases. When we restrict city size to 100,000, the impact is still less than 6%. This seems somewhat puzzling, given that Figure 4 shows that research output in the counterfactual cases is between 5% and 20% lower than in the base case, depending on which scenario one is looking at, for all of the decades of the twentieth century.

The resolution to this puzzle is exactly the negative effect of the technology level $\bar{A}$ on the speed of technological progress. Less effective research effort early in the century leads to a lower level of $\bar{A}$, which in turn makes research later in the century more productive at generating technological progress than in the base case. Consistent with this intuition, the bottom panel of Figure 5 illustrates the results of a “naive” parameterization where $\lambda = 1$ and $\beta = 0$ and each successive unit of research effort yields the same increment to the level of technology. In this case, if city size is restricted to one million, $\bar{A}$ in 2010 is 8% below its baseline level, while if city size is restricted to 100,000, the reduction is about one fifth.

All of the calculations in Figure 5 are based on our preferred value of σ_B (the urban

Figure 5: Counterfactual trajectories of national productivity.



Note: Ratio of counterfactual to observed productivity, $\bar{A}_t^{alt} / \bar{A}_t^{base}$, by decade for two different counterfactuals. City sizes are capped at 1m (squares) and 100k (diamonds). Panels differ in assumptions about the relationship between research output and productivity growth; (a) $\lambda = 0.75, \beta = 2.4$, (b) $\lambda = 1, \beta = 3.1$, (c) $\lambda = 1, \beta = 0$. We assume $\sigma_B = 0.06$.

Table 4: $\bar{A}_{alt} / \bar{A}_{base}$ for $L_{max} = 1,000,000$

Parameters	$\sigma_B = .06$	$\sigma_B = .20$
$\lambda = 1$ and $\beta = 3.1$	0.979	0.935
$\lambda = .75$ and $\beta = 2.4$	0.980	0.939
$\lambda = 1$ and $\beta = 0$	0.924	0.790

Note: Each cell reports $\bar{A}_t^{alt} / \bar{A}_t^{base}$ for 2010 when maximum city size is limited to one million.

scale effect in R&D) of 6%. Table 4 shows the sensitivity of our results to the alternative value of 20% discussed above. We focus on the case where city size in the alternative scenario is limited to one million, and consider the same combinations of λ and β that we consider in Figure 5. To the extent that this effect is relatively small under our baseline parameterization, it would take a very large adjustment of σ_B to produce a large negative effect on productivity.

7. Combining Static and Dynamic Effects

We can now consider the combined effects of reduced aggregate productivity due to slower technological progress ($\bar{A}_t$) and lower static productivity from urban scale effects (the $\tilde{A}_{it}$ s) that would result from a limitation on city sizes. Combining equations (1), (2), (10) and (28) we can write the total change in counterfactual output due to both the types of urban scale effects,

$$\frac{Y_t^{alt}}{Y_t^{base}} = \left(\frac{\bar{A}_t^{alt}}{\bar{A}_t^{base}} \right)^{1/(1-\gamma)} \sum_i \frac{Y_{it}^{base}}{Y_t^{base}} \min \left(1, \left(\frac{L_{max}}{\tilde{L}_{it}} \right)^{\frac{\sigma_A}{1-\gamma} + \sigma_H} \right) \quad (29)$$

Performing the calculation described by equation (29), Table 5 shows output in the alternative case where city size is limited to one million relative to the base case of actual city sizes. We show values for all of the combinations of parameters that we consider above.

For the base-case set of parameters, i.e. $\frac{\sigma_A}{1-\gamma} + \sigma_H = 0.04$, $\sigma_B = .06$, and Bloom *et al.*'s preferred values, output in the alternative case is 8% lower if city sizes are restricted to one million. If we pick values for scale effect parameters at the upper end of the plausible range, $\frac{\sigma_A}{1-\gamma} + \sigma_H = 0.12$ and $\sigma_B = 0.20$, then the reduction in output is 22%. Given the importance often assigned to large cities as drivers of economic growth, these strike us as relatively small effects. For example, on the basis of PPP data from 2022, Canada's per capita GDP is 20% below the US, about the same as in the counterfactual case just considered.

The results in Table 5 can also be transformed to describe the growth rate of output rather than its level. Recall that the experiment we are considering imposes a cap

Table 5: Output Relative to Baseline 2010

$\frac{\sigma_A}{1-\gamma} + \sigma_H$	$\sigma_B = 0.06$			$\sigma_B = 0.20$		
	0.04	0.08	0.12	0.04	0.08	0.12
$\lambda = 1.00, \beta = 3.1$	0.917	0.863	0.816	0.877	0.825	0.780
$\lambda = 0.75, \beta = 2.4$	0.919	0.865	0.817	0.880	0.829	0.783
$\lambda = 1, \beta = 0$	0.866	0.815	0.770	0.741	0.697	0.658

Note: Counterfactual output as a share of realized output in 2010 when counterfactual city size is capped at 1m for different parameter values. Cells in this table are calculated by multiplying the appropriate entries of tables 2 and 4.

on city sizes starting in 1900 – the point in time when our baseline and alternative scenarios diverge. Using data from the Maddison Project, GDP per capita in the United States increased by a factor of 6.1 between 1900 and 2010, corresponding to an annual growth rate of 1.66%. If output in the year 2010 had been 92% of its observed value, corresponding to our base-case parameters, then annual growth would have been 1.58%. Using the parameters representing the upper end of plausible scale effects, the annual growth rate would instead have been 1.43%.

Comparing the last line of Table 5 with the two above shows the importance of the fishing out effect in the Bloom *et al.* production function for technology. When we turn this effect off by setting $\beta = 0$, the decline in output is between 25% and 110% larger. (The relative importance of the fishing-out effect is largest when σ_A is small, so that static scale effects are not important and similarly when σ_B is large, so that scale effects on research productivity are large.)

What could be missing from our analysis that would make the effect of limiting urban scale larger? It is possible that we have simply used incorrect estimates of the scale effects that we include in our model. An alternative is that there are effects of urban scale, either static or dynamic, that we have failed to account for entirely. We can think of several such possibilities.

It may be that there are effects of large cities on human capital that go beyond the effect on wages of individuals who live in or move to those cities. For example, if having New York as a very large city makes everyone in the country have higher human capital, this would be an additional effect. Importantly, we *do* include in our analysis the channel by which New York being a big city affects production of new technology, as proxied by patents. So this would have to be a different type of knowledge production with agglomeration effects that were larger than in the production of patents.

A second possibility is that big cities are important, not because of high population *per-se*, but because having a city be very populous allows a lot of people to take advantage of a particularly good location. This is the improved sorting effect that we discuss in Section 3 but rule out as a path to pursue. Specifically, we assume that the location-time

specific component of productivity, $\hat{A}_{it}$ that each person experiences is the same in the baseline and alternative cases. Indeed, because of difficulties in measurement, we don't even try to estimate how $\hat{A}_{it}$ is related to city size. However, if one took this view, the conclusion would be that big cities are important for growth only because the number of intrinsically good places to locate a city is limited, contradicting the spirit of much of the literature on agglomeration.

A related possibility is that the existence of large cities allows for higher mobility in response to transitory productivity shocks than would be present in the counterfactual case of more smaller cities. For example, if a particular location has a very good productivity shock, many more people can move there in a world of unrestricted city sizes than in a world where sizes are limited.

Finally, it is worth pointing out that our analysis is explicitly about the importance of *big* cities, rather than urbanization in general. Our alternative scenario maintains the same non-urban share of the population as is observed in the baseline.

8. Conclusion

To assess the effect of agglomeration economies on economic growth in the United States, we consider the effect of counterfactual restrictions on city size over the period from 1900 to 2010 on GDP per capita in the year 2010. We allow for both a static effect of city size on productivity and a dynamic effect of city size on research output, which then accumulates over time to determine the level of productive technology.

We conclude that the effects of restricting city size are surprisingly small – or put differently, that there is surprisingly little benefit from agglomeration. To give an example, consider the case in which city size is limited to one million people. In this case, holding the level of technology constant and using our standard set of parameters, we estimate that the loss of the static productivity effect reduces output in 2010 to 94% of its baseline level. In addition, over the 110 year period that we consider, the compounded effect of restricting city size is to reduce the level of technology by 2%, to 98% of its baseline level. Multiplying these effects, 2010 output in the case with limited agglomeration would have been 8% lower than the baseline, and the 1900-2010 GDP growth rate in the alternative scenario would have been less than a tenth of a percentage point lower than the baseline (i.e. 1.58% vs. 1.66%). While this is certainly not a trivial effect, it suggests to us that the urban scale effect was not the primary engine of economic growth.

As with any quantitative conclusion, there are many possible reasons why ours could be wrong. One possibility is that we have incorrectly parameterized either the scale effect of city size on productivity or the similar scale effect on research. But moving to the very

highest end of the range of parameters estimated in the literature does not reverse our finding.

A second possibility is that there are effects of urban scale, either static or dynamic, that we have failed to account for. In the previous section we discussed some of these in detail.

A third possibility is that in examining our particular counterfactual, we have done violence to what people mean when they say that cities are engines of growth. Concretely, we assume the *only* economic effect of limiting city sizes would be via the urban scale effect. A skeptic might point out that if city sizes were limited, there would have to be more cities, and that some of these cities might not have the same fundamental productivity (the term we call $\hat{A}$) as the actually observed cities. This might be due to the new cities not being in locations that are as desirable as the cities that we actually observe. Our answer to this particular critique is that if it is correct, it is not so much cities themselves that are engines of economic growth, but rather good locations in which to put cities.

A final possibility is that we are being too narrow in our interpretation of the phrase “engine of growth.” If urban scale effects explain one-tenth of US economic growth, maybe that qualifies them as being an engine of growth.

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Appendix A. Adjusting the Level of Technology for Urban Scale Economies in Output

This appendix pursues the issue of how taking urban scale effects into account changes the measurement of the speed of technological progress. Literature in this area generally proceeds via the technique of growth accounting, whereby changes in output expected due to observed factor accumulation are subtracted from the observed series of changes in output to derive growth of total factor productivity (TFP). Growth of TFP is in turn assumed to be the same as technological progress. While there are many reasons why there might be slippage between these two concepts, we focus on the effect specific to our paper: If cities are getting larger over time, then urban scale effects will raise measured TFP even in the absence of technological change; or more generally, TFP growth will overstate technological progress. Using the machinery of Section 4, we can correct this problem.

We begin by rewriting the production function, explicitly including the determinants of productivity. For convenience, we also define human capital used in production at the city level as $H_{it} = h_{it}^Y \ell_{it}^Y L_{it}$:

$$Y_{it} = \hat{A}_{it} \bar{A}_t \tilde{A}_{it} K_{it}^\gamma H_{it}^{1-\gamma} \quad (\text{A1})$$

Incorporating urban scale effects, the production function can be rewritten as

$$Y_{it} = \hat{A}_{it} \bar{A}_t \tilde{L}_{it}^{\sigma_A} K_{it}^\gamma H_{it}^{1-\gamma} \quad (\text{A2})$$

We assume that the allocation of factors across cities satisfies

$$(K_i, H_i, L_i) = \alpha_i (\bar{K}, \bar{H}, \bar{L}), \text{ for all } i \text{ and } \alpha_i > 0 \quad (\text{A3})$$

It follows immediately that $\sum_i \alpha_i = n + 1$ and that $\alpha_i = L_i / \bar{L}$.

We can thus rewrite city output yet again as

$$Y_{it} = \hat{A}_{it} \bar{A}_t \tilde{L}_{it}^{\sigma_A} \frac{L_{it}}{\bar{L}_t} \bar{K}^\gamma \bar{H}^{1-\gamma} \quad (\text{A4})$$

Next, let A^* be traditionally measured TFP. The conventional definition for A^* is

$$A_t^* = \frac{\sum_i \hat{A}_{it} \bar{A}_t \tilde{L}_{it}^{\sigma_A} \frac{L_{it}}{\bar{L}_t} \bar{K}^\gamma \bar{H}^{1-\gamma}}{(n+1) \bar{K}^\gamma \bar{H}^{1-\gamma}} = \frac{\bar{A}_t}{(n+1) \bar{L}_t} \sum_i \hat{A}_{it} \bar{A}_t \tilde{L}_{it}^{\sigma_A} L_{it} \quad (\text{A5})$$

This can be rewritten (recalling that $\sum_i \hat{A}_i = n + 1$ requires that $\bar{\hat{A}}_i = 1$):

$$A_t^* = \bar{A}_t \frac{\text{Cov}(\hat{A}_{it}, \tilde{L}_{it}^{\sigma_A} L_{it}) + \overline{\tilde{L}_{it}^{\sigma_A} L_{it}}}{\bar{L}_t} \quad (\text{A6})$$

This can be rewritten as

$$A_t^* = \bar{A}_t \frac{\overline{\tilde{L}_{it}^{\sigma_A} L_{it}}}{\bar{L}_t} \left[1 + \text{Corr}(\hat{A}_{i,t}, \tilde{L}_{it}^{\sigma_A} L_{it}) CV(\hat{A}_{it}) CV(\tilde{L}_{it}^{\sigma_A} L_{it}) \right] \quad (\text{A7})$$

where CV indicates coefficient of variation. The second term in brackets represents the effect on aggregate productivity of the relationship between city size and the component of city productivity that is unaffected by size. It is thus almost certainly positive. Unfortunately, because we can't measure city level productivity $\hat{A}_i$, we have no way of evaluating this term or its changes over time. In calculating technological change, we make the assumption that this term is constant. We can thus derive a simple expression for the relationship between TFP growth and aggregate technological change:

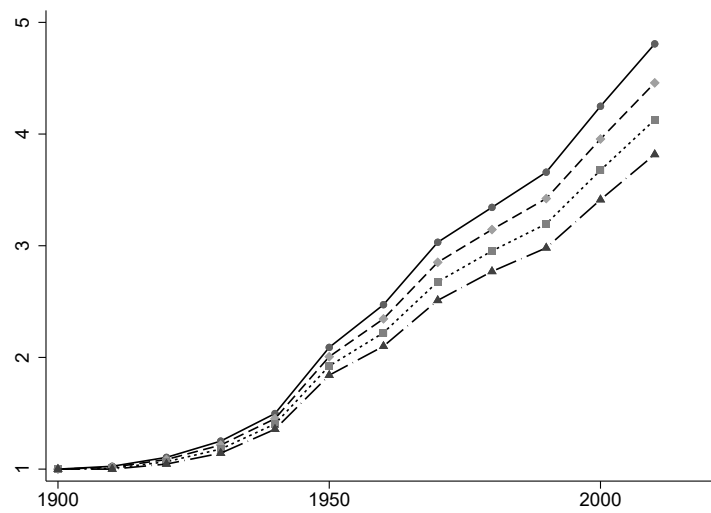
$$\Delta \ln(\bar{A}_t) = \Delta \ln(A_t^*) - \Delta \ln \left(\frac{\overline{\tilde{L}_{it}^{\sigma_A} L_{it}}}{\bar{L}_t} \right) \quad (\text{A8})$$

Equation (A8) shows how to correct aggregate TFP growth to account for increased scale effects over time.

Figure 6 implements equation (A8). The highest, solid black line in the figure reports the uncorrected level of TFP, A^* . Each of the bottom three lines reports a corrected level of technology based on a different estimate of the strength of urban scale economies on output, σ_A . Respectively, the small dashed line, dotted and large dashes describe $\bar{A}_t$ for $\sigma_A = 0.04, 0.08$, and 0.12 .

Unsurprisingly, the gap between the adjusted and unadjusted level of technology grows over the course of the 20th century as cities become larger. By 2010, the adjusted level of technology is 93%, 86% and 79% of its unadjusted level when σ_a is 0.04, 0.08, and 0.12. Alternatively, from 1900 to 2010, the growth rate of the unadjusted level of technology was 1.43% and the corresponding adjusted rates were 1.36%, 1.29%, and 1.22%. Note that these are *not* counterfactual calculations. These are adjustments to the growth rate that result from accounting for changes in the importance of urban scale effects for output.

Figure 6: TFP and Adjusted Technology for Different Urban Scale Effects for Output



Note: Figure shows raw measure of the level of TFP, A_t^* , as the highest line. Three lines below report, from top to bottom, the adjusted level of technology, $\bar{A}_t$, when $\sigma_A = 0.04, 0.08$, and 0.12 .